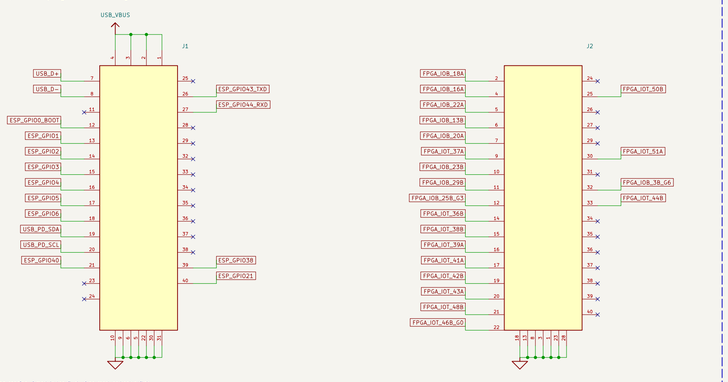
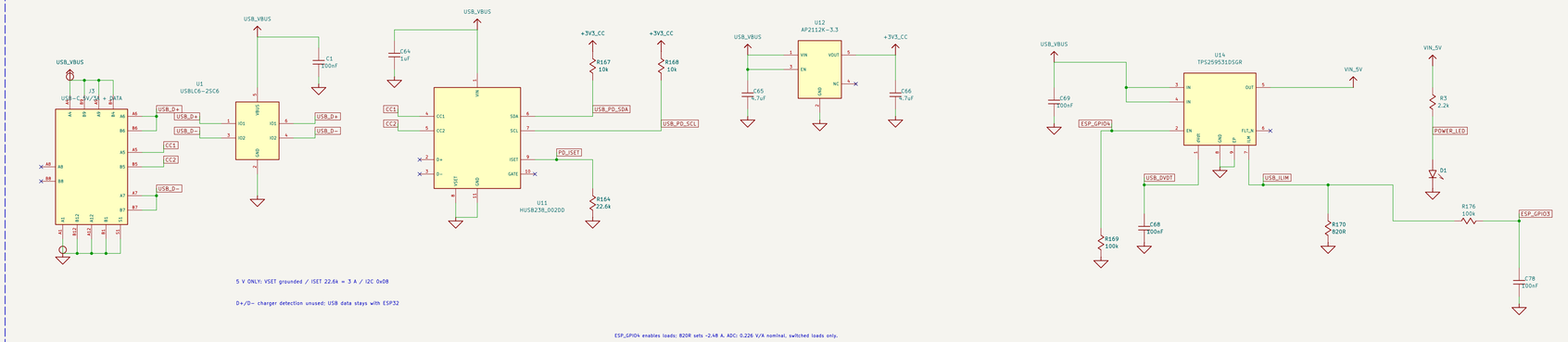
MODULE v1 / exact programmer mating interface



25 mm mating spacing / module top flush / no reset or boot switch

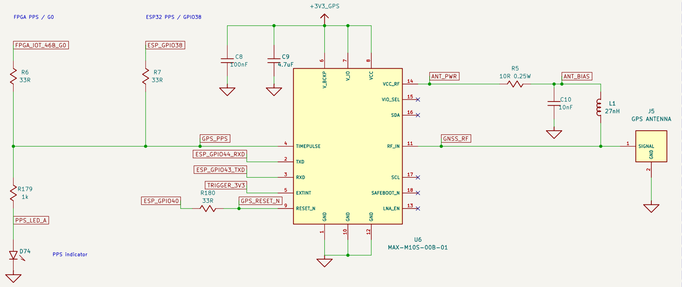
USB-C PD / 5 V, 3 A sink and protected power



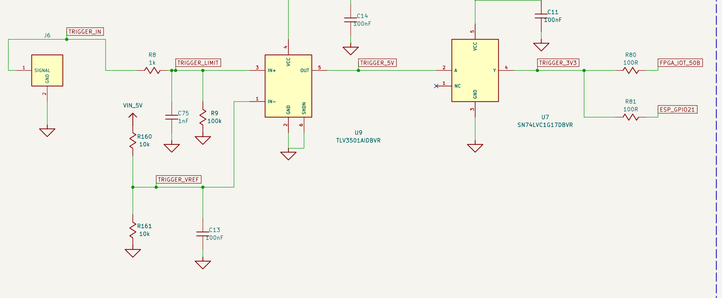
ESP_GPIO4 enables loads: 820R sets ~2.4A A. ADC: 0.226 V/A nominal, switched loads only.

Module and logic boot from raw 5 V. GPIO7/8 read PD status over I2C; firmware budgets current before enabling loads.

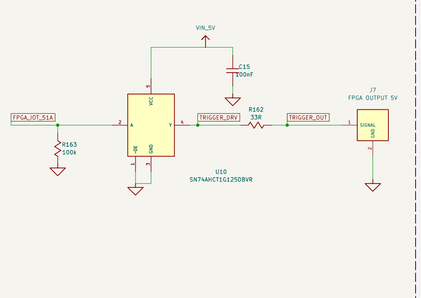
GNSS / PPS reference and active antenna bias



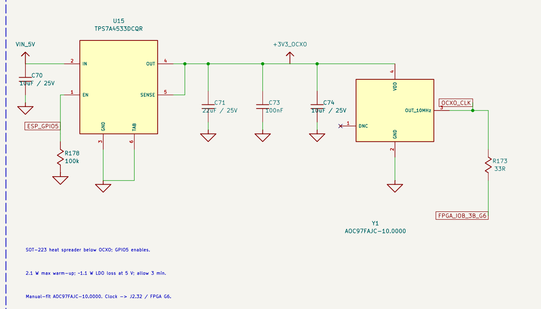
Trigger input / 5 V TTL / threshold ~2.5 V



Trigger output / 5 V TTL



10 MHz OCXO / 3.3 V linear supply

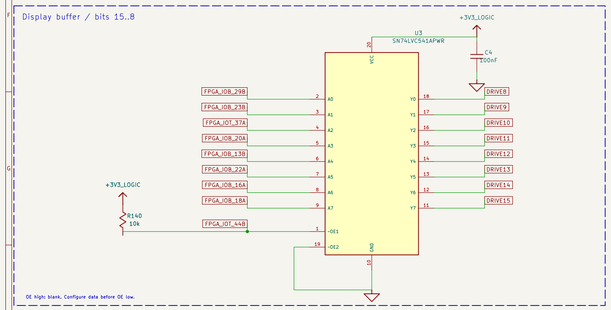


SOT-223 heat spreader below OCXO: GPIO5 enables.

2.1 W max warm-up: ~1.1 W LDO loss at 5 V: allow 3 min.

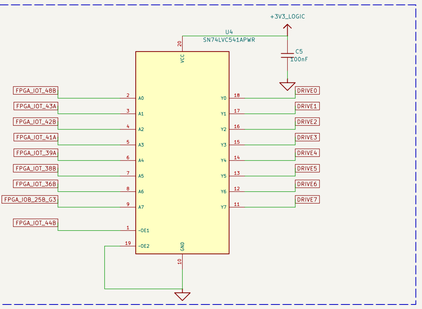
Manual-fit ASDX97FAJC-10.0000. Clock → J2.32 / FPGA G6.

Display buffer / bits 15..8



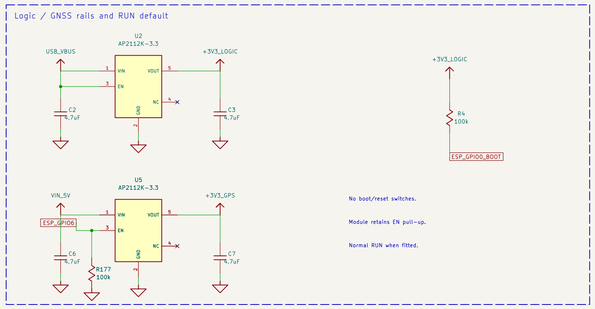
OE high: blank. Configure data before OE low.

Display buffer / bits 7..0



OE high: blank. Configure data before OE low.

Logic / GNSS rails and RUN default

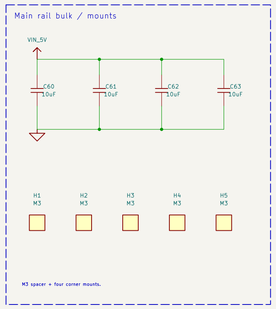


No boot/reset switches.

Module retains EN pull-up.

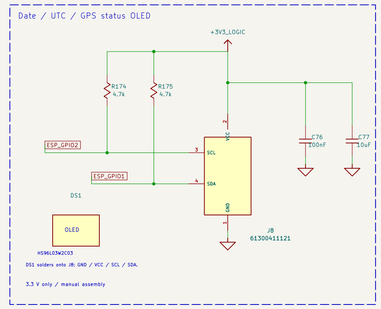
Normal RUN when fitted.

Main rail bulk / mounts



M3 spacer + four corner mounts.

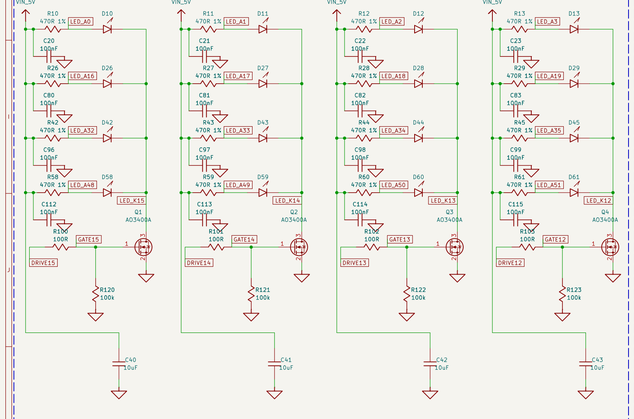
Date / UTC / GPS status OLED



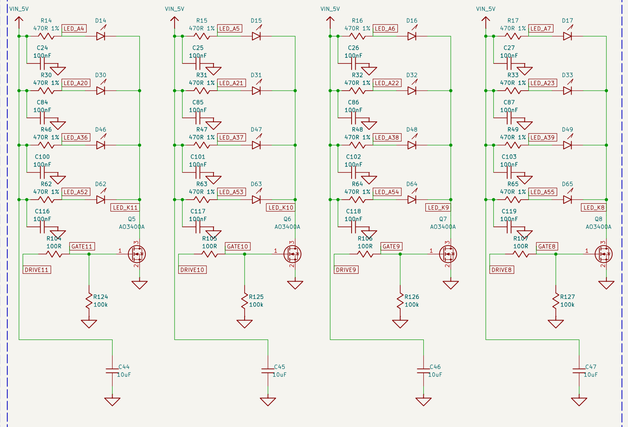
DS1 solders onto J8: GND / VCC / SCL / SDA.

3.3 V only / manual assembly

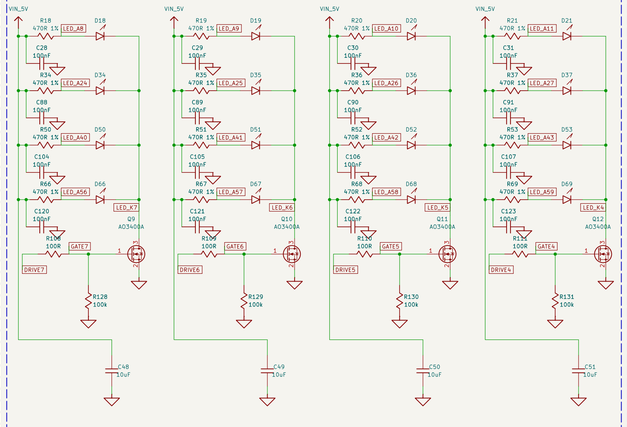
OPTICAL BANK 1 / bits 15..12 / four identical rows



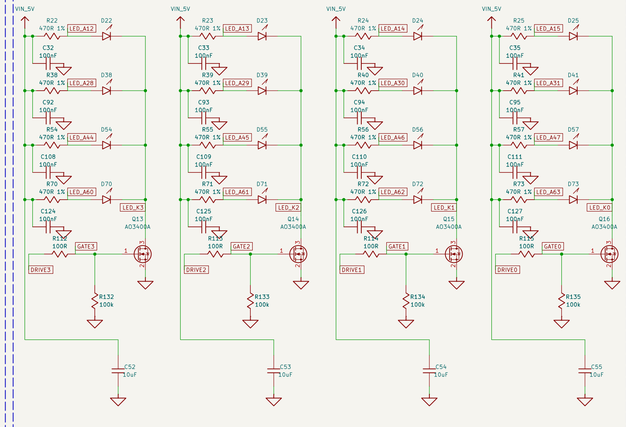
OPTICAL BANK 2 / bits 11..8 / four identical rows



OPTICAL BANK 3 / bits 7..4 / four identical rows



OPTICAL BANK 4 / bits 3..0 / four identical rows



Diagnostic probe pads / functional names on PCB / bare copper, no assembly parts

TP1 / VBUS	TP2 / 5V_LOAD	TP3 / 5V_LOGIC	TP4 / 3V3_CC	TP5 / 3V3_GPS	TP6 / 3V3_OCXO	TP7 / LOAD_EN	TP8 / OCXO_EN	TP9 / GPS_EN	TP10 / PD_SDA	TP11 / PD_SCL	TP12 / I_MON	TP13 / PPS_SRC	TP14 / PPS_FPGA		
VIN_5V	VIN_5V	+3V3_LOGIC	+3V3_CC	+3V3_GPS	+3V3_OCXO	ESP_GPIO5	ESP_GPIO6	ESP_GPIO45	USB_PD_SDA	USB_PD_SCL	ESP_GPIO3	ESP_PPS	FPGA_IOT_48B_G5		
TP15 / PPS_ESP	TP16 / CLK_FPGA	TP17 / 5V_OVP	TP18 / 5V_CMP	TP19 / 3V3_5V	TP20 / 5V_PG	TP21 / OUT_FPGA	TP22 / OUT_DRV	TP23 / GPS_TX	TP24 / GPS_RX	TP25 / GPS_RST_N	TP26 / ANT_PWR	TP27 / ANT_BIAS	TP28 / LED_DC_N	TP29 / OLED_SCL	
ESP_GPIO38	FPGA_IOB_3B_G6	TRIGGER_VREF	TRIGGER_5V	TRIGGER_3V3	FPGA_IOT_51A	TRIGGER_DRV	TRIGGER_OUT	ESP_GPIO44_RXD	ESP_GPIO43_TXD	GPS_RESET_N	ANT_PWR	ANT_BIAS	FPGA_IOT_44B	ESP_GPIO2	
TP30 / SDA_FPGA	TP31 / SCL_FPGA														
FPGA_IOT_46B_G0	FPGA_IOT_50B														

bp-test
Sheet: /
File: calibrator.kicad_sch

Title: GPS optical time calibrator / module v1 carrier

Size: A1	Date:	Rev: A draft
KiCad E.D.A. 10.0.6		Id: 1/1